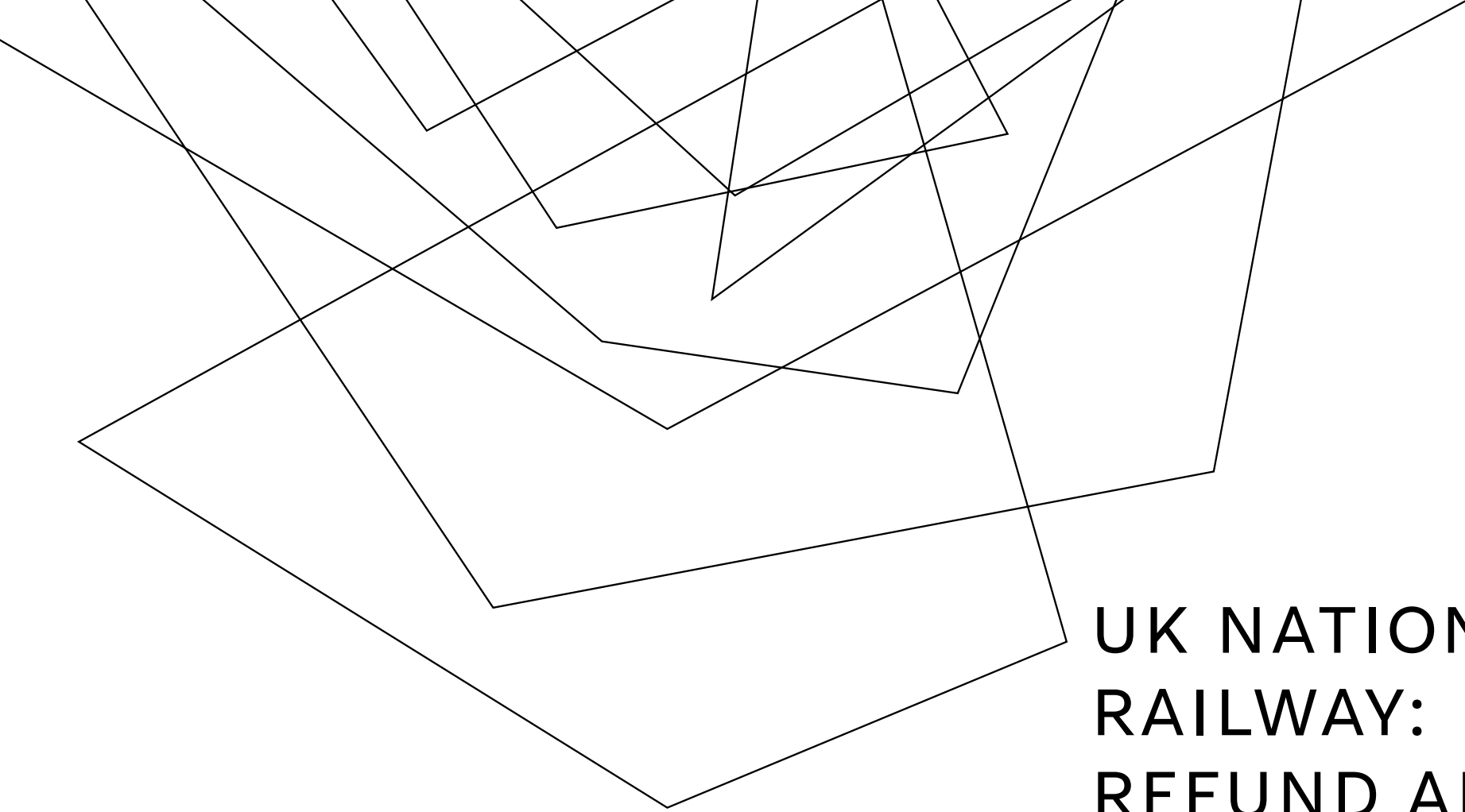


Abstract geometric lines in black on a white background, forming various overlapping polygons and shapes, primarily concentrated on the left side of the page.

UK NATIONAL RAILWAY: REFUND ANALYSIS

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INTRODUCTION

Railways process millions of ticket transactions daily, with refund requests impacting customer satisfaction and revenue. This analysis examines refund patterns and their underlying reasons using MS Excel. Key insights reveal why refund requests are high and which routes require early intervention to improve service.

BUSINESS OBJECTIVE

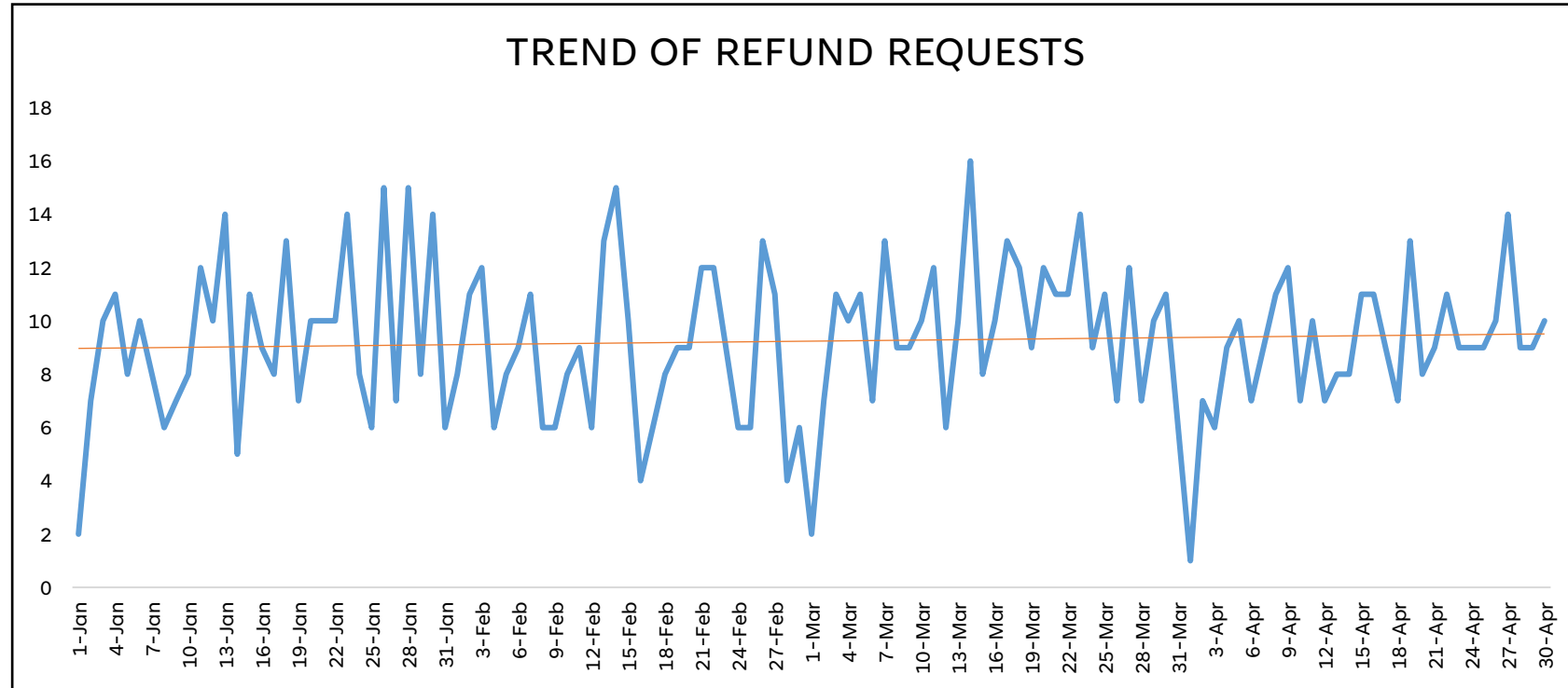
Identify top reasons and routes leading to refund requests and recommend actions to reduce refund requests and financial losses.

DATASET OVERVIEW

Source: Maven analytics (mock railway transactional data for **UK National Rail**) from December 2023 to April 2024.

This dataset provides a comprehensive view of railway operations and customer behavior. It includes detailed information on customer segmentation, travel routes, ticket types, pricing, journey dates and times, as well as refund requests and their underlying reasons. It offers a realistic representation of UK National Rail operations, enabling in-depth analysis of ticket sales, refund patterns and passenger trends.

INSIGHTS

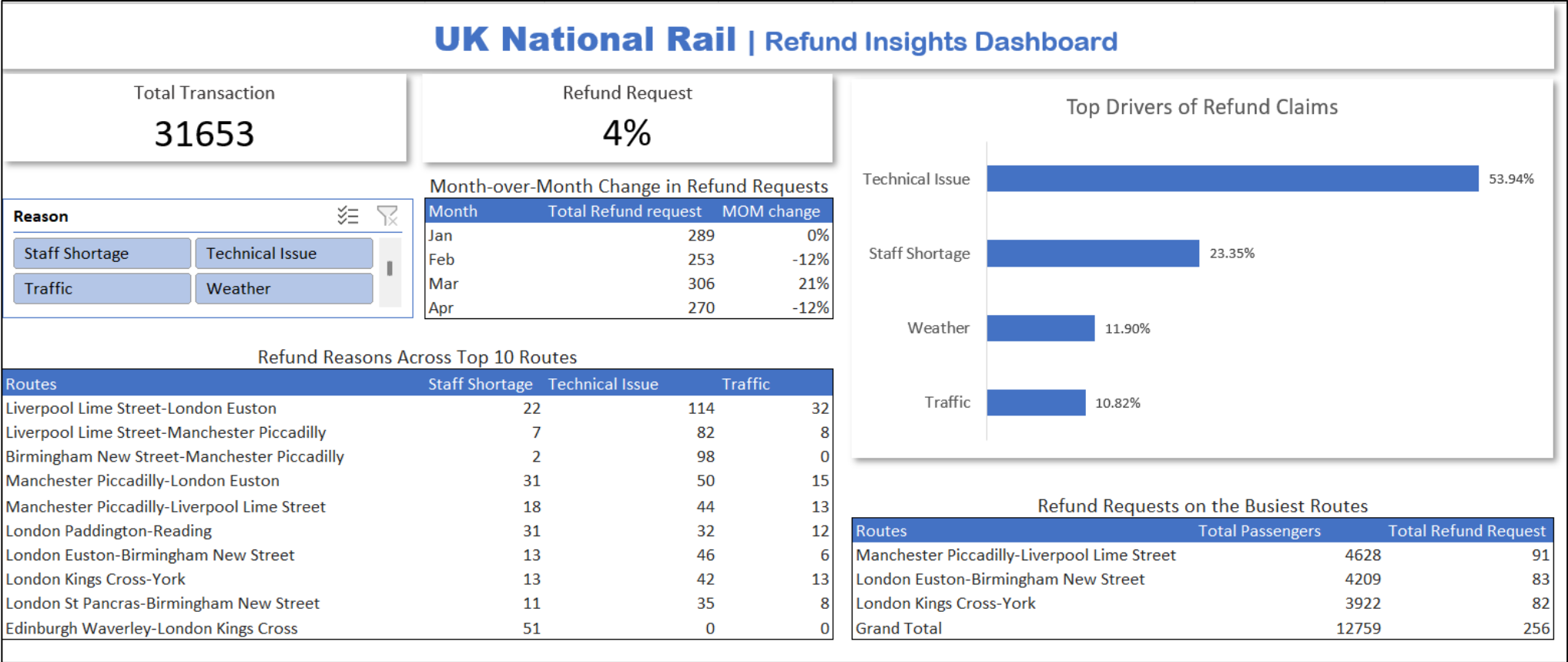


The trend line shows that refund requests rate have remained constant for months, suggesting that no effective measures have been implemented to bring down refund requests.

INSIGHTS

- A total of **31,653 transactions** were recorded, with only **4% resulting in refund requests**.
- The **top reasons for delays** were: *technical issues, staff shortages and weather conditions*.
- Refund requests **declined in February**, rose again in March, and later returned to the same percentage.
- Across stations, **technical issues consistently ranked as the leading cause** of refunds.
- Routes with the highest technical issues:
 - Liverpool Lime Street – London Euston
 - Liverpool Lime Street – Manchester Piccadilly
 - Birmingham New Street – Manchester Piccadilly
- Stations like **Liverpool Lime Street, London Euston, and Manchester Piccadilly** appeared repeatedly in the top 10 routes, indicating concentrated technical issues at these locations.
- **Manchester Piccadilly – Liverpool Lime Street** emerged as the busiest route in the UK.
- The **Edinburgh Waverley – London King's Cross** route was mainly affected by *staff shortages*.

DASHBOARD OVERVIEW



RECOMMENDATIONS

- **Address recurring technical issues** at major stations (Liverpool Lime Street, London Euston, Manchester Piccadilly) through maintenance and infrastructure upgrades.
- **Increase staff availability**, especially on routes like Edinburgh Waverley – London King's Cross where staff shortages dominate.
- Monitor and act on **seasonal spikes** in refund request (e.g., March).
- **Enhance service reliability on busiest routes** (Manchester Piccadilly – Liverpool Lime Street) to reduce refund rates and protect revenue.
- **Develop targeted communication strategies** to update passengers on technical issues and minimize dissatisfaction leading to refund claims.

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THANK YOU